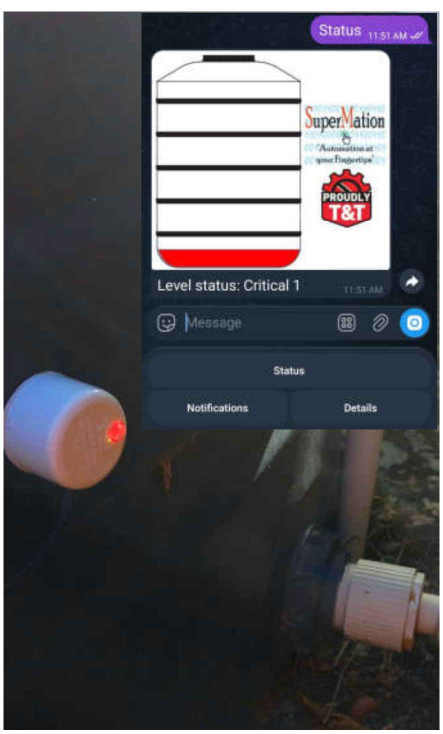
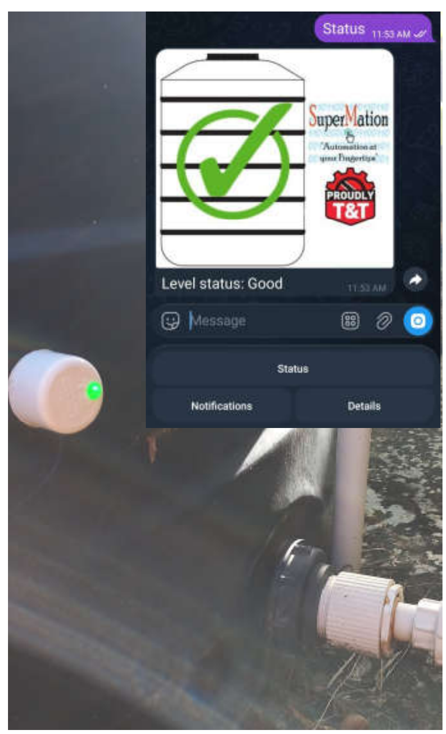
TankMatic™ L1

Installation and Operation Manual

Non-Intrusive ‘Dry Run’ Pump Protector (supports up to 7 tanks) with Remote Access





Date: 7 April 2026

Contents

List of Abbreviations

- SPL - SuperMation Products Limited.
- UI - User Interface.

Models Applicable:

- L1.

1 Prerequisites

Safety is crucial so please ensure proper safety rules and use of tools are followed. The details of these are outside the scope of this document. SPL is not liable for any injuries or damages incurred due to incorrect installation.

1.1 Tools

Before installation, ensure the following tools are available:

1. Hot glue gun (minimum 60Watt) for full size (11mm / 0.43inch) sticks.
2. Extension cord (length depends on the location of the target tank and nearest power receptacle).
3. Ladder or scaffolding for tanks installed at significant height.
4. Clean, lint-free, cloth and rubbing alcohol in a spray bottle.

1.2 Site resources

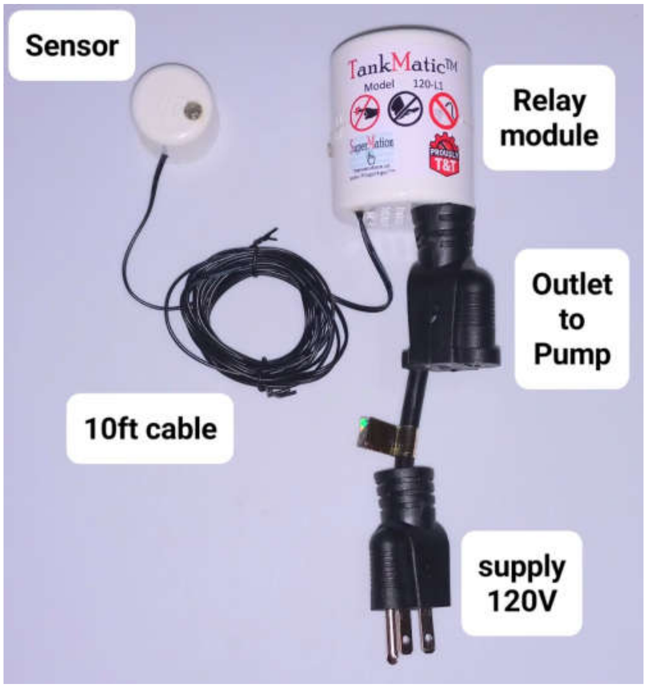
The following are needed for the installation at the tank:

1. One available 120VAC 3 pin NEMA power receptacle. This is used by the water pump.
2. WiFi 2.4GHz network (signal strength should be -70dBm or stronger). Internet for remote access and notifications.
3. The tank outer surface must be dry with no trace of moisture. Please do not install in drizzle/rain.

1.3 Kit contents

The kit contains all that is needed for a full installation as listed below:

1. One TankMatic L1 relay module.
2. One TankMatic L1 sensor module. External sensor pads (up to 6) are dependent on the client's order.
3. One power cord 10 foot long for sensor.
4. One hot glue stick 11mm x 270mm.
5. One nameplate sticker.


Sensor
Relay module
Outlet to Pump
10ft cable
supply 120V
TankMatic™
Model 120-L1

2 Connect the product to the WiFi network

2.1 Power up

Insert the 3 prong power plug into a 120VAC receptacle, preferably within 20 feet of the WiFi router.

2.2 Enter WiFi credentials

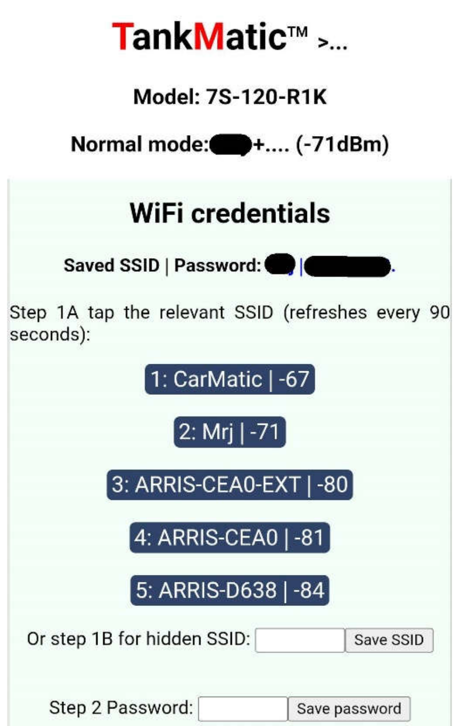
The Sensor will flash purple for approximately 30 seconds then flash cyan for 120 seconds. This pattern will repeat until it successfully connects to a router. The cyan flashes indicate it is in configuration mode. Use your phone to connect to its private network:

SSID: TM1 (the number may be different if multiple products are installed).

Password: 1234567890

When connected use the UI URL in the receipt email or scan the below QR code (this works only when connected to the R model private network):

The UI will load, tap on the About tab then scroll down to the WiFi credentials section:



There are 5 buttons showing the discovered SSIDs followed by their signal strength in dBm (the strongest signal is at button 1). Tap the button of the relevant SSID or enter your hidden SSID manually, tap the Save SSID button, then tap the back button on your phone. Enter the relevant password then tap the Save password button. The product will restart into normal mode. It will flash purple several times. If the credentials are correct and the router accepts the product, the light will turn yellow or green.

If the product did not connect successfully, repeat this sub-section (3.2) and carefully re-enter the credentials. If the saved SSID and password are correct and connection is not successful, please check the WiFi router settings.

2.3 Find the product's profile in Telegram

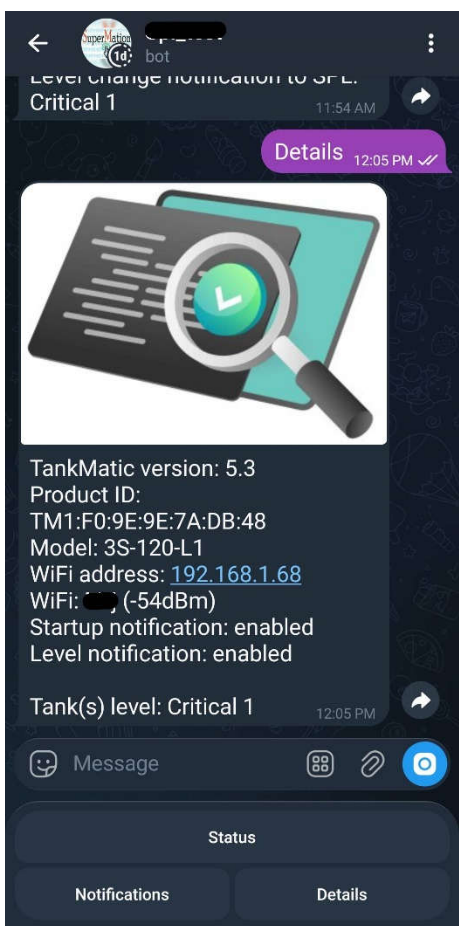
Use the provided URL in the receipt email to add the product in Telegram. Tap the Start button and the keyboard menu will now be visible. Do not share this information with unauthorized individuals.

The product must be used at least once every 6 months or else the profile is remove by the Telegram network. The sales agreement does not cover reprogramming of the product.

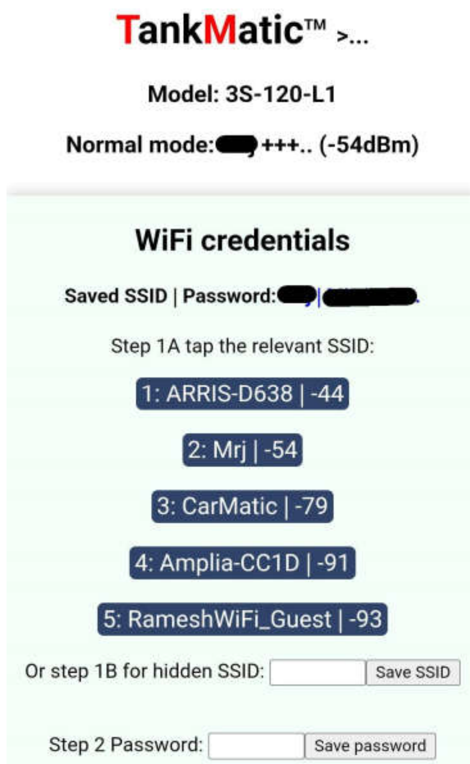
2.4 Notifications

2.4.1 Add or remover Users

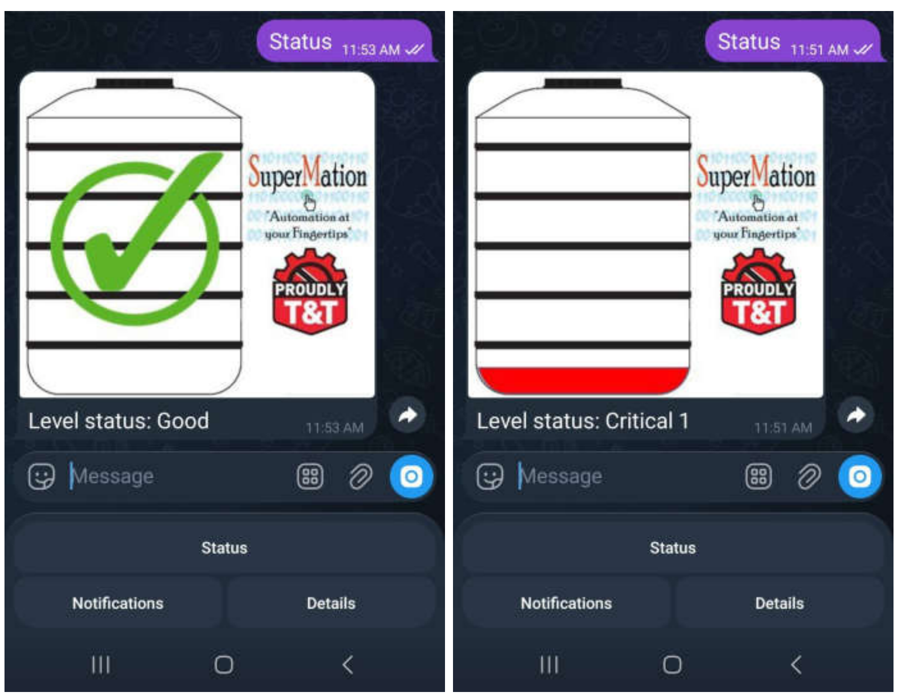
While connected to the WiFi router, use the UI URL in the receipt email. If using an Android phone, you may need to wait approximately 5 minutes before this address resolves. A faster method is to tap the Details button in the Telegram chat keyboard: the address will be shown in blue text (see example below). Please bookmark both the UI URL and numerical IP addresses in your browser app.



In the detailed UI (see sub-section 3.2), in the Control tab scroll down to the Notifications via Telegram section.



In Telegram tap the Status button, wait for a response (as shown in the below pictures) and tap User 1 store button.



A second user can be added by sharing the Telegram program, tapping the Status button and tapping User 2 store button.

Either user can be removed by tapping the erase buttons and typing Yes at the prompt.

2.4.2 Enable or disable notifications

In sub-section 3.4.1, tap on the Notify On/Off buttons to enable or disable the startup and level change notifications. Both are enabled by default. Also in the Telegram chat, tap on the Notifications buttons to perform the same functions.

3 Installation onto the Tank

The product is designed for plastic (polypropylene) water storage tanks. Ceramic, concrete, metal and silicone tank walls are not compatible with the product. The maximum power ratings for 1-phase pump motors are shown below:

Voltage (V)	Horsepower (HP)	Kilowatt (kW)
110 to 120	1.25	0.93
220 to 240	2.5	1.86

The motor nameplate running current must not exceed 11 Amps.

3.1 Powering and Testing

The 120VAC plug needs to be inserted into the power receptacle used by the water pump and ensure the LEDs are active. The water pump will plug into the receptacle at the relay module.

The sensor LED will be red prior to installation and the relay LED will flash once every 6 seconds. To test the unit, press firmly the sensor base (opposite side of the LED) to anywhere on the tank below the water level and the LED will be green. For the first power-up of the product, the relay module LED will go solid blue and an audible click will be heard: the pump may activate depending on the discharge line pressure.

3.2 Mark and clean the external wall

Identify the sensor installation location(s) then clean using rubbing alcohol. Ideally at least 12 inches from the base of the tank to ensure a reserve volume of water is always available to the pump.

3.3 Install the components

Ensure your hot glue gun is hot enough that some glue dribbles from the nozzle. Use hot glue to secure the relay module to the power receptacle housing.

Apply a ring of glue directly behind the sensor (and a square of glue behind sensor pads if ordered) and press onto the tank at the location(s) in sub-section 2.2.

If the installation was incorrectly done, spray the cooled glue with rubbing alcohol and let soak for 2 minutes. Pull off the component and peel off traces of the glue. Repeat the installation properly.

The above picture shows a typical installation of the relay module glued onto the pump's power receptacle.

The above pictures show the sensor (without external sensor pads) on a tank. The left picture indicates no water at the senor, the right picture shows water has risen above the sensor.

4 Calibration and General Use

4.1 Calibration

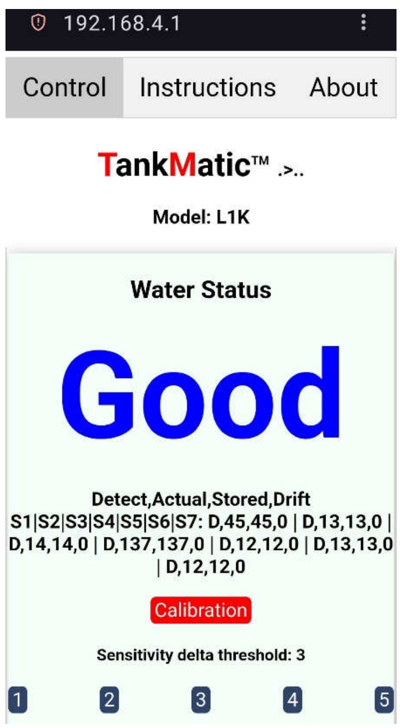
Calibration is recommended for reliable operation. To perform a calibration ensure the water level is above the sensor, wait 30 seconds after power-up and connect to the product's WiFi network, for example:

SSID: TM1 (the number may be different if multiple products are installed).

Password: 1234567890

To perform a calibration with custom sensitivity, access its UI on your phone's browser app. In your browser app address bar enter the follow local address: 192.168.4.1 or scan the below QR code.

The UI will load as shown below for a tank with water above the sensor.



The individual sensors' data is shown: D for detected and – for no detection. The product has one internal sensor and supports 6 external sensor pads each with a 10 foot long cable. Distances between the tanks will determine the feasibility of multiple sensors.

Tap the Calibration button, type Yes and tap OK. The sensor LED will flash white. It is now calibrated and the relay will activate immediately if this was the first power-up.

The sensitivity delta threshold is default at 3. This is suitable for most polypropylene tanks. Smaller values can be used for thicker walled tanks, typically very tall tanks. Higher values can be applied to thin walled tanks.

4.2 Use

The product can be accessed:

1. Locally via its UI on your phone browser app
2. Remotely using the Telegram app.
3. Remotely using the data stream. This is for clients with a custom monitoring system and is useful for fleet management of tanks.

The product LEDs has colour codes for the sensor and relay modules.

Sensor	
Green	Water detected at the sensor(s). If external sensor pads are installed, water must be present at all sensors.
Red	At least one sensor is above the water level. Use the UI to determine which sensor(s) have not detected water.
White flash	Calibration in progress. Use only when the water level is above the sensor.
Yellow flash	Phone or tablet has just connected to the product.
Relay	
Rapid rate flashing	Power-up has occurred and the relay is searching for the sensor.
Medium rate flashing	The relay is reconnecting to the sensor.
One flash every 6 seconds	No water detected and the relay has turned off the pump.
Solid	Water detected and the relay has enabled the pump. The motor will run depending on the pump's pressure switch.
Two flash every 6 seconds	5 minute safety delay-on when the water level has risen above the sensor. After this delay the LED will go solid and enable the pump. To bypass this time delay (not recommended), unplug the relay module then plug again into the receptacle. The product will power up and enable the pump immediately if the water level is above the sensor.

4.3 Troubleshooting

High pressure water jets onto the tank may cause a false positive indication if the tank's water is below the sensor(s). When the tank outer surface dries, the product will properly detect the internal level.

If the relay LED is solid colour and the pump is not running, turn on a water faucet to allow the pump pressure switch to power the motor. If the pump does not run, contact SPL.

On a known full tank of water, if the sensor shows red, perform calibration as sub-section 3.1. The LED should be green. If it is not green, check the adhesion of the sensor(s) onto the tank wall. If adhesion is good and the LED is not green, contact SPL.

If the product keeps going into configuration mode (cyan flashes) after successful installation, please check the WiFi router settings (some routers limit the number of connected clients). So long as the saved SSID password are correct, there is no need to re-enter the WiFi credentials; the product will reconnect when the router allows it. Typically a power cycle will solve most issues.